

GOOGLE SUMMER OF CODE 2026

Improve Getting Started Documentation

A structured onboarding proposal for dora-rs, adora, and practical embodied-AI examples.

PROJECT PROPOSAL BY

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PROJECT PERIOD

May - August 2026

APPLICANT

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SUBMISSION DATE

March 2026

CONTENTS

Table of Contents

A compact map of the proposal, from motivation to deliverables.

1. Project Overview	3
1.1 Background	3
1.2 Objectives	3
2. Applicant Strengths	3
3. Technology Stack	4
4. Timeline	5
4.1 Community Bonding Period	5
4.2 Coding Phase 1	6
4.3 Coding Phase 2	7
5. Detailed Work Plan	7
5.1 Pre-Midterm Goals	7
5.2 Pre-Final Goals	8
6. Deliverables	8
7. Risk Assessment	9
8. Reference Links	9
Commitment	10

01

Project Overview

A beginner-centered documentation effort for dora-rs and adora.

1.1 Background

dora-rs is a high-performance dataflow framework for building robotic applications. As the project evolves, the learning curve for new users has become a bottleneck for community growth. This project aims to create systematic getting-started documentation, video tutorials, and example code to lower the entry barrier and help developers quickly grasp the core concepts and usage of dora-rs and adora.

1.2 Objectives

Core objective

Create a comprehensive learning system for dora-rs and adora projects, including documentation tutorials, video tutorials, and runnable example code, enabling new users to complete the journey from installation to running their first complete example within 2-3 hours.

Specific objectives include:

- Write clear, step-by-step getting-started documentation.
- Produce accompanying video tutorials for Bilibili and YouTube.
- Provide complete, runnable example code.
- Demonstrate practical use cases combining dora-rs with AI tools.

2. Applicant Strengths

I have made substantial contributions to the dora-rs ecosystem prior to this application, with the following advantages:

02

Applicant Strengths

Existing contributions and the technical foundation for the proposal.

Contribution History

Project	Contribution	Link
dora-examples	Example code contributions	https://gitcode.com/dora-org/dora-examples
dora-rs-org	Project website contributions	https://github.com/dora-rs-org/dora-rs-org
Video Tutorial	Beginner-friendly tutorial	https://www.bilibili.com/video/BV1gzzHBCEFn

Video Tutorial Highlights

- Based on the latest Windows 11 environment.
- Demonstrates using AI tools, including Claude Code, for programming and debugging.
- Starts from scratch and remains suitable for complete beginners.

3. Technology Stack

This project involves the following technologies and tools:

Technology / Tool	Purpose
dora-rs	Core robotic dataflow framework
adora	High-level wrapper for dora-rs
rerun	Visualization and simulation tool
Python	Primary development language
Rust	Performance-critical modules
PowerShell	Windows environment scripting
Claude Code	AI-assisted programming tool

03

Timeline

A staged plan from community bonding to final submission.

4.1 Community Bonding Period (May 1 - May 24)

Time	Activities
May 1-7	Establish contact with mentors, clarify project details and expectations.
May 8-14	Deep dive into dora-rs and adora documentation, and understand latest developments.
May 15-21	Start handling small contributions and become familiar with the code submission workflow.
May 22-24	Develop a detailed Phase 1 work plan.

4.2 Coding Phase 1 (May 25 - July 10)

Coding officially begins: May 25, 2026

04

Coding Phase 1

Documentation foundations, first examples, and midterm preparation.

Week	Date Range	Main Tasks
Week 1	May 25-31	dora-rs and adora introduction docs, installation guide, and Hello World example.
Week 2	June 1-7	rerun simulation tool introduction, create first simple scene.
Week 3	June 8-14	Claude Code usage tutorial, write example code driving rerun scene.
Week 4	June 15-21	Retrieve camera data from rerun, implement visual feedback.
Week 5	June 22-28	Consolidate previous content, refine documentation and videos.
Week 6	June 29 - July 5	Midterm evaluation preparation, submit work products.

Midterm evaluation deadline: July 10, 2026 18:00 UTC

05

Coding Phase 2

From multimodal perception to embodied task planning examples.

Week	Date Range	Main Tasks
Week 7	July 11-17	Integrate open-source multimodal models and analyze scene visual information.
Week 8	July 18-24	Acquire more sensor data and expand scene interaction capabilities.
Week 9	July 25-31	Plan action paths using LLM large models.
Week 10	August 1-7	Integrate agentsdk for automated task planning.
Week 11	August 8-14	adora + octos advanced integration, creating the ultimate intelligent experience project.
Week 12	August 15-21	Final consolidation, complete all documentation and videos.
Final Week	August 22-24	Submit final work product and mentor evaluation.

Final submission deadline: August 24, 2026 18:00 UTC

5. Detailed Work Plan

5.1 Pre-Midterm Goals

1. **dora-rs and adora Introduction**

Write complete installation guide covering Windows, Linux, and macOS platforms. Create Hello World example demonstrating basic dataflow node communication.

2. **rerun Simulation Introduction**

Introduce basic concepts and usage of rerun visualization tool. Create a simple simulation scene, such as a vehicle or robotic arm, as foundation for subsequent tutorials.

06

Pre-Final Goals

Completing the learning path and publishing usable materials.

3. Claude Code Usage Tutorial

Write tutorial on using Claude Code for dora-rs development. Create example code demonstrating AI-driven vehicle or arm in rerun scene.

4. Camera Data Acquisition

Implement camera data retrieval from rerun scene. Demonstrate integrating visual data into dora-rs dataflow.

5.2 Pre-Final Goals

5. Multimodal Model Integration

Integrate open-source multimodal models, such as LLaVA and Qwen-VL, to analyze scene visual information. Expand sensor data acquisition capabilities, including depth and pose.

6. LLM Action Planning

Leverage large language models for task understanding and action planning. Implement conversion from natural language instructions to robot actions.

7. agentsdk Automated Task Planning

Integrate agentsdk framework for automated task decomposition and execution. Create automated planning examples for multi-step tasks.

8. adora + octos Advanced Integration

Explore deep integration of adora with octos for more intelligent robot control. Create ultimate intelligent experience demonstration project.

6. Deliverables

All deliverables will be published in the following formats:

07

Deliverables and Risk Assessment

Expected outputs and practical mitigation plans.

Type	Platform	Content
Documentation	GitHub / GitCode	Complete tutorial documentation in Markdown format.
Example Code	GitHub / GitCode	Runnable code for each milestone.
Video Tutorial	Bilibili	Chinese video tutorials accompanying documentation.
Video Tutorial	YouTube	English video tutorials for international users.

7. Risk Assessment

Risk	Likelihood	Mitigation
Technical dependency updates causing compatibility issues	Medium	Regularly sync upstream updates and maintain communication with mentors.
Multimodal model integration complexity exceeds expectations	Medium	Prepare simplified alternatives and prioritize core functionality.
Insufficient time for video production	Low	Prioritize documentation quality; videos can be released in phases.
Personal time conflicts	Low	Plan ahead and sync progress with mentors weekly.

8. Reference Links

- dora-examples project: <https://gitcode.com/dora-org/dora-examples>
- dora-rs-org project: <https://github.com/dora-rs-org/dora-rs-org>
- Getting Started Video Tutorial: <https://www.bilibili.com/video/BV1gzzHBCEFn>
- GSoC 2026 Official Timeline: <https://summerofcode.withgoogle.com/>

08

Commitment

A clear commitment to delivery, communication, and quality.

I commit to completing this project according to the above plan, maintaining regular communication with my mentor, and ensuring the quality of deliverables. Any changes to the plan will be discussed with my mentor in advance and approved.

Applicant

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Project Focus

Accessible documentation, runnable examples, video tutorials, and beginner-friendly embodied-AI learning paths for dora-rs and adora.